Luis Carlos Macêdo Moreira

Backend Software Engineer · Python (Django, DRF, FastAPI) · AWS Cloud · Applied AI (LLMs, RAG, agents) · React/TypeScript

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Summary

Backend software engineer with 1.5 years in production across 3 companies, focused on Python (Django, DRF, FastAPI) and asynchronous architecture with Celery and Redis. I've built services that deal with slow third-party APIs, millions of records and concurrency, deployed on AWS (S3, Lambda). I build the AI layer (LLMs, RAG with pgvector, agents with human-in-the-loop) and the React frontend when the product needs it. I work with tests (170+ written), CI, code review and technical documentation, on remote teams in Brazil and the United Kingdom.

Technical skills

- **Backend and data:** Python, Django, Django REST Framework, FastAPI, Celery (+ Beat), Redis, PostgreSQL, pgvector, query optimization
- **Cloud and infrastructure:** AWS (S3, Lambda), Docker, GitHub Actions (CI/CD)
- **Architecture:** asynchronous and distributed processing with queues and workers, multi-tenancy, idempotency, retries, RBAC, JWT, OAuth 2.0
- **Applied AI:** LLMs (Gemini, Anthropic, LiteLLM), RAG with pgvector, agents with human-in-the-loop
- **Frontend:** React, TypeScript, Vite, Tailwind CSS
- **Quality and collaboration:** Git, GitHub, pull requests, code review, pytest, mocks, technical documentation

Experience

São Luís City Hall (SEMAD) — Backend/Full Stack Developer

May 2025 – present · systems in production

- **Slaughterhouse Sanitary Inspection (DRF + PostgreSQL + React):** lead developer, end to end. I turned a manual process digital, with digitally signed reports and an audit trail.
- Replaced a fragile authentication scheme with JWT and dynamic RBAC (role × resource × action); migrated the backend from FastAPI to DRF and set up CI with tests and deploys.
- **HR reports (Django + Celery + Redis):** the state HR API was too slow for real-time use. I built scheduled, idempotent syncs into PostgreSQL and reports generated in the background.

Dizevolv — Software Engineer

August – October 2026 · remote · multi-tenant legal and real estate SaaS built from scratch

- **Legal platform with LLMs and RAG (Django + Celery + pgvector):** orchestrated searches against a third-party API with polling, retries and quota control. A recovery task re-enqueues stuck jobs when a worker goes down.

- **AI assistants on WhatsApp:** customer service with LLMs (Gemini and Anthropic via LiteLLM) and human supervision.
- **Real estate CRM (DRF + React):** idempotent endpoints via Idempotency-Key, with transaction.atomic and select_for_update to prevent race conditions.

CodeLeap — Backend Developer

October 2025 – June 2026 · London, United Kingdom (remote) · mature Django codebases for international products

- **Eliminated 504 timeouts:** moved exports of millions of records to Celery, with keyset pagination, CSV on S3 and a link sent by email. Replaced N+1 queries with annotate using Exists/Subquery and Prefetch.
- **OAuth integrations (Google and Microsoft) for an AI assistant:** single-use tickets in Redis, consumed atomically with a Lua script.
- LLM-based code grader running on AWS Lambda.
- 170+ tests (Django TestCase + mocks), including regression suites, within a pull request and code review workflow.

Education, languages and projects

- **UNDB** — B.Sc. in Software Engineering · Aug 2022 – Dec 2026
- **English:** full professional proficiency (C1, EF SET 70/100) · Portuguese: native
- **Autoescola Bless landing page** (React + Tailwind), live in production: autoescola-bless.com.br